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CRICKET PLAYER PERFORMANCE TRACKING APPLICATION : CRICKSTATS

Submitted

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Thank You

CRICKSTATS

Cricket Player Performance Tracking Application

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1. Abstract

Cricket is one of the most popular sports in the world, especially in countries like India, Australia, England, and South Africa. Performance analysis plays a crucial role in modern cricket because teams rely heavily on statistics and data to evaluate player performance. Traditionally, cricket statistics were maintained manually using score sheets and notebooks. However, manual tracking of player performance is time-consuming and prone to errors.

The CRICKSTATS application is designed to track and analyze the performance of cricket players efficiently using a digital system. This application allows coaches, teams, and individual players to store and monitor performance metrics such as runs scored, strike rate, batting average, wickets taken, bowling economy, and match history.

The main objective of this system is to simplify the process of maintaining cricket statistics and provide analytical insights into player performance. The system will store match data, calculate statistics automatically, and

generate reports for analysis.

By using CRICKSTATS, users can easily access player performance history, compare statistics, and make better decisions regarding player selection and training. The application helps in improving performance monitoring and provides a systematic approach to sports analytics.

2. Introduction

Cricket is a game where performance evaluation is extremely important. Every player is judged based on their statistics such as runs, wickets, batting average, strike rate, and consistency in matches.

With the advancement of technology, sports analytics has become an important part of modern cricket. Professional teams use advanced systems to track player performance and improve strategies. However, local teams, academies, and amateur players often lack access to such systems.

CRICKSTATS is a software application designed to bridge this gap by providing a simple yet effective platform for tracking cricket player performance. This application allows users to record match data and automatically calculate player statistics.

The system provides a structured way of managing cricket data. Instead of maintaining records manually, users can enter match details into the application, and the system will process the information and generate statistical reports.

The application focuses on improving accuracy, saving time, and providing better insights into player performance. It also helps coaches analyze players and identify strengths and weaknesses.

3. Problem Statement

In traditional cricket management systems, player statistics are often maintained manually using paper records or simple spreadsheets. This approach has several limitations.

Manual data entry increases the chances of errors and makes it difficult to maintain consistency in records. It also requires a significant amount of time to calculate averages and analyze performance trends.

Another major issue is the lack of centralized data storage. Player information and match records may be stored in different places, making it difficult to retrieve historical data when needed.

Coaches and team managers require accurate statistics to evaluate players and make decisions regarding team

selection and training strategies. Without a proper system, analyzing performance becomes difficult.

Therefore, there is a need for a dedicated application that can efficiently store cricket statistics, calculate performance metrics automatically, and provide easy access to data.

The CRICKSTATS system is designed to solve these problems by providing a digital platform for managing cricket performance data.

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4. Objectives of the Project

The main objectives of the CRICKSTATS application are:

1. To develop a digital platform for tracking cricket player performance.
2. To maintain a centralized database of player statistics.
3. To automate the calculation of batting and bowling statistics.
4. To provide match history and player performance reports.
5. To assist coaches and players in analyzing performance.
6. To reduce manual effort and improve accuracy in record keeping.
7. To create a user-friendly interface for easy data entry and retrieval.

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5. Scope of the Project

The CRICKSTATS application is designed primarily for cricket academies, local teams, and amateur players who want to maintain performance statistics.

The system allows users to:

- Store player profiles
- Record match details
- Track batting and bowling statistics
- Generate performance reports
- Compare players based on statistics

The scope of the project is limited to performance tracking and statistical analysis. It does not include live match

streaming or advanced AI-based prediction systems.

However, the system can be extended in the future to include features such as real-time match scoring, player ranking systems, and advanced analytics.

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6. Literature Survey

Sports analytics has become an essential component of modern sports management. Many professional cricket teams use specialized software to analyze player performance.

Applications such as Cricbuzz, ESPN Cricinfo, and other sports analytics platforms provide detailed statistics about international matches and players. However, these platforms mainly focus on professional cricket and do not allow local teams to manage their own data.

Research studies show that performance analysis helps teams improve strategies and identify player strengths and weaknesses. By analyzing past performances, teams can make informed decisions regarding player selection and training.

Several sports management systems have been developed to track statistics in different sports such as football, basketball, and baseball. These systems use databases and automated calculations to generate performance reports.

The CRICKSTATS application is inspired by these systems and aims to provide a simplified version suitable for academies and local teams.

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7. Existing System

In the existing system, cricket statistics are usually recorded manually using notebooks or spreadsheets.

Limitations:

- Time-consuming process
- High chances of calculation errors
- Difficult to maintain large amounts of data
- No automated analysis
- Difficult to retrieve past records

Because of these limitations, managing player performance data becomes inefficient.

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8. Proposed System

The proposed system is the CRICKSTATS application, which provides a digital solution for tracking cricket player performance.

Features:

- Player profile management
- Match record storage
- Automatic calculation of statistics
- Performance reports
- Data analysis tools

Advantages:

- Improved accuracy
- Faster data processing
- Easy access to player statistics
- Better performance evaluation

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9. System Architecture

The system architecture consists of three main components:

1. **User Interface Layer**
   Allows users to interact with the application and enter match data.
2. **Application Logic Layer**
   Processes the data and calculates statistics such as batting average and bowling economy.
3. **Database Layer**
   Stores player profiles, match records, and statistical data.

The architecture ensures efficient data flow between the user interface and the database.

10. Modules of the System

Main modules:

1. **Player Management Module**
   Stores player details such as name, age, role, and team.
2. **Match Management Module**
   Records match information such as date, teams, and scores.
3. **Performance Tracking Module**
   Calculates runs, wickets, averages, strike rates, etc.
4. **Statistics Analysis Module**
   Generates performance reports and comparisons.
5. **User Management Module**
   Handles login and access control.

11. System Requirements

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: 4 GB
- Hard Disk: 500 GB

Software Requirements

- Operating System: Windows / Linux
- Programming Language: Java / Python / PHP
- Database: MySQL
- IDE: Visual Studio Code / Eclipse

12. Database Design

Main tables:

1. Players
2. Matches
3. Batting_Stats
4. Bowling_Stats
5. Users

Each table stores specific information required for the system.

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13. Data Flow Diagram (DFD)

Level 0 DFD shows interaction between user and system.

User inputs:

- Player data
- Match details

System outputs:

- Player statistics
- Performance reports

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14. Use Case Diagram

Actors:

- Admin
- Coach
- Player

Use cases:

- Add player
- Record match

- View statistics
- Generate reports

15. Implementation

The CRICKSTATS system is implemented using a programming language and database system. The user interface allows easy data entry and viewing of statistics.

The backend processes data and performs calculations for player performance metrics.

16. Advantages

- Accurate performance tracking
- Easy data management
- Automated statistical calculations
- Time saving
- Better decision making

17. Limitations

- Requires digital device access
- Initial setup needed
- Limited to cricket statistics only

18. Future Enhancements

Future improvements may include:

- Mobile application version
- Real-time match scoring

- AI-based performance prediction
- Cloud data storage
- Player ranking system

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19. Conclusion

The CRICKSTATS application provides a simple and efficient solution for tracking cricket player performance. It eliminates the need for manual record keeping and provides accurate statistical analysis.

The system helps players and coaches understand performance trends and make better decisions regarding training and team selection.

With further development, the application can become a powerful tool for sports analytics and cricket performance management.

THANK YOU ....